

NAME	: MR. Kiran K T	VISIT ID	: 25030156584
GENDER	: Male	AGE	: 26 Years
DATE OF REGISTRATION	: 16-July-2025 05:18	DATE OF COLLECTION	: 16-July-2025 05:18
REFERRED BY	:	DATE OF REPORT	: 16-July-2025 10:08
REF CENTER	: KARUNYA CLINICAL LABORATORY	REF NO.	:

LABORATORY TEST REPORT				
TEST PARAMETER	RESULT	UNIT	REFERENCE RANGE	SAMPLE TYPE

CLINICAL BIOCHEMISTRY				
T3 -TOTAL	1.32	ng/ml	0.80 - 2.0	Serum
Method : ECLIA				
T4 -TOTAL	8.65	ug/dl	5.1 - 14.1	
Method : ECLIA				
TSH (Ultra Sensitive)	2.54	μIU/mL	0.4 - 4.2	
Method : ECLIA				

Limitations: T3 and T4 circulates in reversibly bound form with Thyroid binding globulins (TBG), and to a lesser extent albumin and Thyroid binding Pre Albumin, so conditions in which TBG and protein levels alter such as pregnancy, excess estrogens, androgens, steroids may falsely affect the T3 and T4 levels. Normal levels of T4 can also be seen in Hyperthyroid patients with T3 Thyrotoxicosis, hypoproteinemia or ingestion of certain drugs. Serum T4 levels in neonates and infants are higher than values in the normal adult, due to the increased concentration of TBG in neonate serum. TSH may be normal in central hypothyroidism, recent rapid correction of hyperthyroidism or hypothyroidism, pregnancy, phenytoin therapy. Autoimmune disorders may produce spurious results. Various drugs can interfere with the test result. TSH has a diurnal rhythm so values may vary if sample collection is done at different times of the day.

Interpretation(s):
TSH stimulates the production and secretion of the metabolically active thyroid hormones, thyroxine (T4) and triiodothyronine (T3), by interacting with a specific receptor on the thyroid cell surface. The synthesis and secretion of TSH is stimulated by Thyrotropin-releasing hormone (TRH), the hypothalamic tripeptide, in response to low levels of circulating thyroid hormones. Elevated levels of T3 and T4 suppress the production of TSH via a classic negative feedback mechanism. Failure at any level of regulation of the hypothalamic-pituitary-thyroid axis will result in either underproduction (hypothyroidism) or overproduction (hyperthyroidism) of T4 and/or T3.

Clinical Condition	T3 Levels	T4 Levels	TSH Levels
Primary hypothyroidism	Reduced	Reduced	Incresed (Significantly)
Subclinical hypothyrodism	Normal or low Normal	Normal or low Normal	High
Primary hyperthyrodism	Increased	Increased	Reduced (at times undetectable)



DR.CHETHANA CHETHAN MBBS M.D., Consultant
Biochemist
KMC Reg. No: 41296